Gating time constants are the conserved coordinates of cardiac ion channels: a unified reanalysis of seven voltage-clamp corpora

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Abstract

Kinetic models of ion channels are written as rate-constant schemes, and the schemes are known to be unidentifiable from voltage-clamp data. What the data do identify has not been mapped at scale. We first show on a calibrated thirteen-parameter hERG model that the identified subspace is seven-dimensional, that the time scale is pinned by the waveform while amplitude is an exact gauge direction, and that the identifiable content is therefore the set of modal coordinates: time constants, steady-state curves and delays. We then measured those coordinates directly. A single extraction pipeline, with per-row profile-likelihood confidence intervals, was applied uniformly to seven public voltage-clamp corpora, yielding a master table of 487,584 measured quantities from 20,709 cells covering 74 channel types in seven families. The intervals are calibrated (96.2% coverage at the nominal 95% level). On this table, one pre-registered verdict separates the two kinds of quantity. Time constants are conserved across cells while voltage midpoints drift: the midpoint scatter exceeds the time-constant scatter in 63 of 64 activation groups (median ratio 2.47), in deactivation, inactivation and recovery groups, in all five HCN groups with reversed gating polarity, in nine of ten layers of 933 published models, and in 38 cross-species pairs where time constants stay within two-fold while midpoints move by millivolts. Temperature separates the classes again: activation carries the pooled Q_{10} of 3.78 against 1.99 for deactivation, and dividing 25 °C tables by pooled Q_{10} values predicts 35 °C kinetics with 94% of 325 points inside a two-fold window. Across a five-laboratory blinded hERG drug dataset reanalysed at waveform level, drugs move occupancy and leave deactivation kinetics unchanged (median ratio 0.95, flat against block depth, blinded cells reproducing the result). Models assembled from the measured tables with one free conductance per cell predict held-out CiPA drug-screening traces in 67 of 68 cells. Channel identity is written in kinetics; cell state is written in midpoints.

1 Introduction

Kinetic models of cardiac ion channels are written as Hodgkin–Huxley or Markov schemes and calibrated by global optimisation [1, 2, 3, 4, 5]. The schemes differ in topology but share a parameterisation: one rate constant per transition, chosen so that the model reproduces the recordings. The weakness of that parameterisation is documented. In a survey of thirteen published Markov models of cardiac channels, nine could not be determined uniquely from step-clamp data, and the authors drew the consequence explicitly: where parameters are unidentifiable, the layout of a model cannot be established from the data, and the fitted values do not carry the mechanistic meaning usually attributed to them [6]. The responses since have been richer protocols [7] and added constraints or priors [8]. Both leave the rate constants in place. What has not been tested systematically is whether the data identify a different class of quantity that can be measured directly and compared across cells, channels, laboratories and species.

We reached the same conclusion by a different route. The question is not academic. Cardiac safety pharmacology now runs on in silico models: the CiPA initiative scores drug risk through

channel models and a ventricular cell model, and every regulatory submission inherits the identifiability problem underneath [1, 3]. If the rate constants cannot be recovered, a safety pipeline should rest on what can be recovered.

Our own hERG model was a thirteen-parameter Markov-type scheme built from the published four-state ring with a double-exponential activation edge and a parallel slow component. It fitted the recordings and passed seven of eight pre-set criteria; it failed the eighth on the sinusoidal protocol, after four rounds in which fixing the peak broke the waveform and fixing the waveform broke the peak. Adding the states that would repair the exclusion produced rate multipliers whose Cramér–Rao bounds exceed the data window by a factor of about 760. The two failures are one failure: expressiveness is bought with directions the data cannot see.

A Fisher analysis of the fitted model explains which directions those are, and it points at what can be measured instead. We therefore rebuilt the analysis from the measurement side. Seven public voltage-clamp corpora were re-extracted from raw waveforms by one pipeline: local fits per protocol segment, no global objective, and a profile-likelihood confidence interval on every extracted quantity. The result is a single master table in which a hERG deactivation time constant from a manual patch rig and a Kv1.1 activation time constant from a high-throughput robot carry the same columns and the same uncertainty semantics. On that table we ask four questions in sequence. Which quantities are shared across cells? Does temperature move them? Do drugs move them? And do models built from the shared quantities predict data they never saw?

The answer is consistent enough to state up front. Time constants are shared; voltage midpoints are cell-specific. Temperature moves time constants by a process-specific Q_{10} and leaves midpoints flat. Drugs move occupancy and leave kinetics essentially unchanged. And models assembled from measured tables with a single free conductance per cell forward-predict held-out protocols, held-out cells and held-out laboratories.

2 Results

2.1 The identified coordinates are modal

The starting point is a concrete failure. The thirteen-parameter hERG model reproduces the four protocol families of its construction cell and passes seven of eight pre-set criteria. The eighth fails informatively: the peak current and the sinusoidal waveform cannot be satisfied by the same parameter set, the exclusion persists across four re-parameterisation rounds, and the residual of 0.00096 in R^2 is an order of magnitude above run-to-run scatter while the noise ceiling of the dataset sits at $R^2 \geq 0.973$. The residual is model error, not benchmark artefact. Adding serial closed states repairs the waveform and forfeits the parameters: the repaired variant passes all eight criteria, but its rate multipliers carry Cramér–Rao bounds of ±9.6–13.2 dex against a data window of ±0.017 dex.

The Fisher information of the calibrated thirteen-parameter hERG model,

$$F_{ij} = \frac{1}{\sigma^2} \sum_t \frac{\partial I(t;\boldsymbol{\theta})}{\partial \theta_i} \frac{\partial I(t;\boldsymbol{\theta})}{\partial \theta_j}, \tag{1}$$

was evaluated by central finite differences on an independent simulation channel with per-segment residual variance as weights. The thirteen eigenvalues span 12.9 orders of magnitude. Six directions sit below 10^{-3} of the largest, so the data identify a seven-dimensional subspace of the thirteen-dimensional parameter space. The null space is structured. One degeneracy is exact: with the maximum open probability frozen, the combination $\alpha_{\text{amp}} \cdot \text{ALPHA} \cdot \text{XMAX}$ enters only as a product, and no experiment can separate its factors. One is near-exact: the two exponential components of the activation edge can be exchanged at a cost consistent with machine precision ($\lambda/\lambda_{\max} \approx 1.4 \times 10^{-13}$).

The degeneracy usually assumed in this literature is not there. A joint rescaling of all rate constants by s with the conductance by $1/s$ is often treated as leaving voltage-clamp protocols

invariant; it does not. On a fixed protocol time grid,

$$I_s(V,t) = \frac{1}{s}\, I(V, st) \neq I(V,t), \tag{2}$$

and a 10% rescaling changes the predicted waveform by 18–45%, far outside the noise. The data pin the time scale. What they do not pin is the amplitude scale: because the current factorises as

$$I(t) = g\, \Phi(t; \boldsymbol{\theta}), \tag{3}$$

with g the conductance and Φ the shape, the Fisher information about the shape parameters obeys $F_{\boldsymbol{\theta\theta}} = g^2 F_0$, independent of g. The shape subspace and the scale direction decouple exactly. Amplitude is a gauge direction; time is not. The identified coordinates of a gating model are therefore the modal ones: how fast the current rises and falls at each voltage, where it sits at steady state, how long opening is delayed. These are observables of protocol-rich recordings, not fitting targets.

2.2 One measurement pipeline over seven corpora

We applied one extraction pipeline uniformly to seven public voltage-clamp corpora (Table 1): the Channelpedia automated-patch collection (17,987 cells after quality control, 66 Kv-family channel types plus HCN, Kir, K2P and other families), the five-laboratory blinded hERG drug dataset coordinated by HESI under the FDA Broad Agency Announcement (28 drugs, 1,875 cells passing trace-level quality control), the five-temperature hERG series of Lei and colleagues (670 wells), the nine-cell manual hERG dataset of Beattie and colleagues, a Nav1.5 whole-cell series at two temperatures, a $Ca_v1.2$ series with continuous bath-temperature readout, and an I_{Ks} activation series (283 recordings). Every protocol segment was reduced locally: mono- or bi-exponential fits to activation, inactivation, deactivation and tail segments; Boltzmann fits to steady-state curves; delays by back-extrapolation. No objective function couples protocols or parameters at any stage.

The corpora differ in structure, and the pipeline meets each on its own terms. For the CiPA dataset the sweep metadata exist for only one laboratory, so vehicle, drug-block and steady-state windows were segmented from the waveform plateaus alone, voltage-contaminated current columns in three laboratories were detected and excluded, and current units were read from column suffixes (the two official numerical disasters and the unit pitfalls found in the public tables are listed in the Supplementary Information). For the nine-cell manual dataset, steady-state activation was measured by the tail-current method rather than the holding-segment mean, which removes a Boltzmann degeneracy that would otherwise push the midpoint fits to their bounds. For Nav1.5 the inactivation time constant was measured from peak-aligned decay windows at 40 µs sampling. For $Ca_v1.2$ the bath-temperature channel was carried alongside each sweep, so temperature enters as a measured covariate rather than a nominal label. For I_{Ks} the activation delay and time constant were fit jointly on the step epoch and the deactivation time constant on the −40 mV tail epoch, with duplicate recordings of the same cell resolved to a single best version before fitting.

Each extracted quantity carries a per-row profile-likelihood 95% confidence interval rather than a symmetric standard error. The intervals matter for two reasons. They are honest where the likelihood is asymmetric (the median upper to lower half-width ratio across the Channelpedia corpus is 1.43, so symmetric errors systematically understate the upward uncertainty), and they are calibrated: across the 255,411 quality-passed time-constant rows of the master table, the nominal 95% intervals contain the point estimate at 96.2%, with per-source coverage between 93.4% and 100% (Fig. 1c). Rows whose fit touches a window boundary are flagged rather than deleted (for example the Ba^{2+}-carried $Ca_v1.2$ records, where inactivation is too slow for the 180 ms window), and they enter every statistic with their flag.

The merged table holds 487,584 measured quantities from 20,709 cells across 74 channel types in seven families, with source, cell, protocol, value, interval, quality flag and method version on every row (Fig. 1a,b). Three reference layers are stored in the same schema: the official CiPA

potency anchors for 29 drugs across seven channels, 38 cross-species group comparisons, and ten gating-layer summaries of 933 published models from the ICGenealogy model library, refit in-house. All analysis below reads this table and nothing else.

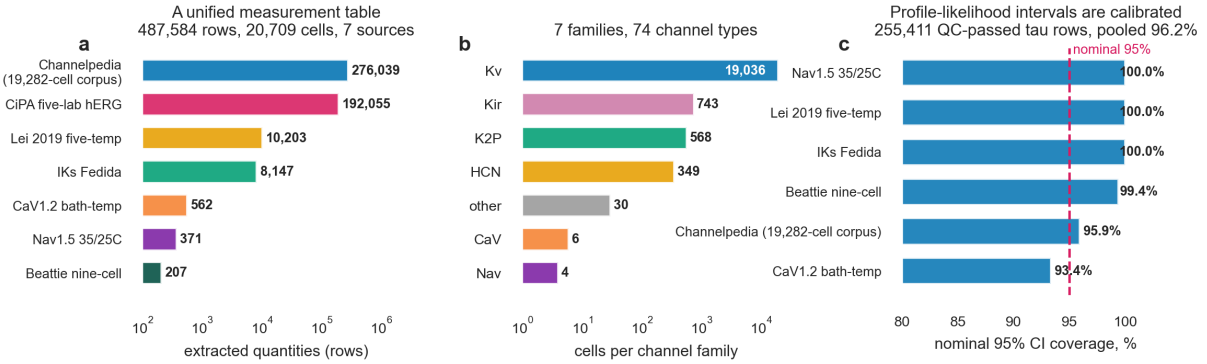


Figure 1: **One extraction pipeline, one table.** (a) Measured quantities per corpus (rows of the master table). (b) Cells per channel family. (c) Calibration of the per-row profile-likelihood intervals: coverage of the point estimate by the nominal 95% interval, per corpus (quality-passed time-constant rows only; pooled coverage 96.2%, $n = 255{,}411$).

2.3 Time constants are conserved, midpoints are not

For every group of cells of one channel type at one temperature we computed two scatters: the scatter of the time constant across cells at fixed voltage, in dex, and the scatter of the steady-state midpoint across cells, in units of the Boltzmann slope k (so that both are dimensionless measures of relative dispersion). The verdict rule is the ratio of the two, with group-level shrinkage by hierarchical partial pooling and per-row uncertainties propagated from the profile intervals. A ratio above one means the time constant is the more conserved quantity.

The verdict is nearly unanimous, and it is the same verdict at every level of organisation (Fig. 2a). In the Channelpedia corpus the main activation judgement passes in 63 of 64 pure-Kv groups (98.4%), with median ratio 2.47. The same rule holds for deactivation (43 of 48, median 2.82), inactivation (28 of 35, median 1.33) and recovery from inactivation (39 of 42, median 2.15). The HCN family, whose gating polarity is reversed, passes all five of its groups with median ratio 8.03, and its subtype midpoint ordering (HCN1 > HCN2 > HCN4 > HCN3) reproduces the literature, a lateral check that the extractor reads both directions of gating correctly. The rule also holds in the published-model layer: across 933 deduplicated models from ICGenealogy, whose gating curves we refit individually, nine of ten family-by-gate layers pass (median 2.07), the exception being the small and highly inbred I_H model set, which ties at 1.00. Finally, between species, the time constant stays within two-fold in 38 of 38 cross-species channel pairs while midpoints differ by up to tens of millivolts (Fig. 2c).

The single failure of the main verdict is registered rather than smoothed over. Kv3.2 at 15 °C returns a ratio of 0.83: its kinetic scatter is among the largest in the corpus and its midpoint scatter the smallest, so this channel at this temperature stores its diversity in kinetics, as its source study also notes. We also tested the window-current hypothesis, that activation and inactivation midpoints of one cell are jointly constrained; it is negative (median correlation 0.30 across 35 channel-temperature pairs), so the two midpoints are independent set points of the cell.

The verdict is stable under four changes of calibre. Removing the Boltzmann fits that hit their voltage bounds leaves 63 of 64 groups passing (median 2.87). Replacing the two-stage variance decomposition by full hierarchical partial pooling shrinks both sides and leaves the verdict unchanged (63 of 64, median 2.48). Replacing symmetric standard errors by the profile-likelihood intervals changes exactly one of the 194 group verdicts across the five tables (Kv1.7 at 15 °C moves

from 1.009 to 0.995, sitting on the threshold in both calibres) and leaves the main verdict at 63 of 64 with median 2.47. And the hierarchical time-constant estimates differ from the frozen values by a median of −1.3% per group, so the scatter estimates are nearly unbiased. The 66-group frame of earlier drafts contained two TASK-3 groups; K2P leak channels have no sigmoidal gating and were registered as a grouping impurity, giving the 64-group frame used here.

Fig. 2b shows the raw geometry behind one verdict group. In Kv1.1 at 25 °C the activation time constant at fixed voltage clusters within 0.58 dex across 592 cells, while the activation midpoint spreads over 8.65 slope units. The conservation statement is therefore not about measurement precision but about what is variable between cells: the cell sets its operating point, and it inherits its kinetics.

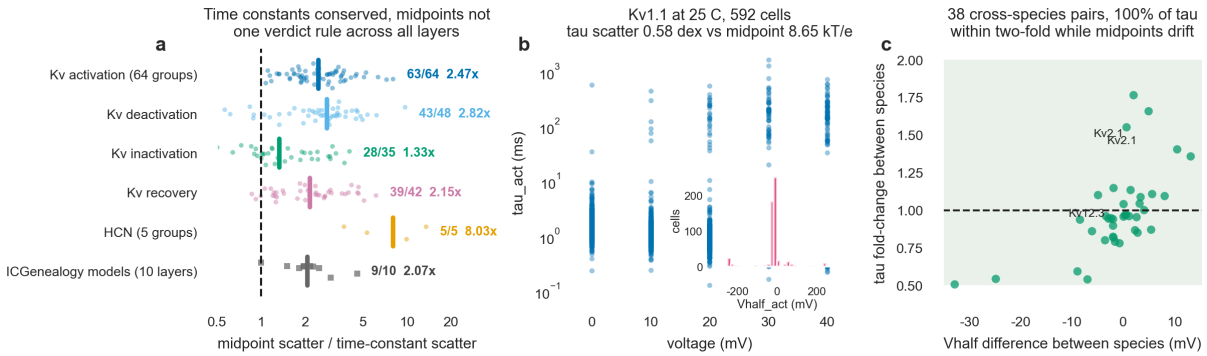


Figure 2: **One verdict rule across all layers.** (a) Ratio of midpoint scatter to time-constant scatter per group, for five experimental layers and the published-model layer; points are groups, bars are medians, and the dashed line marks equality. Counts and medians use the shrunk (partial-pooling) estimates with profile-interval uncertainties. (b) One group in raw form: Kv1.1 activation time constants at 25 °C versus voltage across 592 cells; inset, the midpoint distribution of the same cells. (c) Cross-species pairs: time-constant fold-change against midpoint difference for 38 pairs; the band marks two-fold.

2.4 Temperature separates the two classes and carries a forecast

If kinetics are channel identity and midpoints are cell state, temperature should move the first in an organised way and need not move the second at all. That is what the table shows. In the five-temperature hERG series (670 wells, 25–37 °C), the recovery time constant follows $Q_{10} = 2.91$, the fast activation component 2.37 and the fast deactivation component 2.79, each monotone across the five temperatures (Fig. 3a). In the Channelpedia corpus, where many channel types were recorded at more than one temperature, the pooled Q_{10} is 3.78 for activation ($n = 202$ channel-process pairs), 1.99 for deactivation ($n = 74$) and 1.84 for inactivation ($n = 65$): temperature loads on activation across the family (Fig. 3b). Midpoints, in contrast, are flat in temperature at the family level (median −0.05 mV per °C); the two bidirectional midpoint drifts reported for hERG [16, 17] are hERG-specific, not a general rule.

The $Ca_v1.2$ corpus adds a continuous readout: bath temperature was recorded on a third channel during pharmacology ramps, and the calcium-dependent inactivation time constant, extracted per sweep, decreases gently with temperature (apparent $Q_{10} \approx 1.4$ over 32–38.5 °C; Fig. 3d), consistent with a process carried more by calcium binding than by a protein conformational change.

Temperature then gives the strongest test available to a measurement table: a zero-parameter forecast. Dividing the 25 °C population time-constant tables by the pooled per-channel, per-process, per-voltage Q_{10} predicts the 35 °C population tables with no free parameter. Of 325 predicted points across 20 channel types, 94% land within a two-fold window of the measured 35

°C values and the median fold error is 28% (Fig. 3c). The prediction errors are not random: they concentrate where the 25 °C population is thin, which is a sampling limit rather than a method limit.

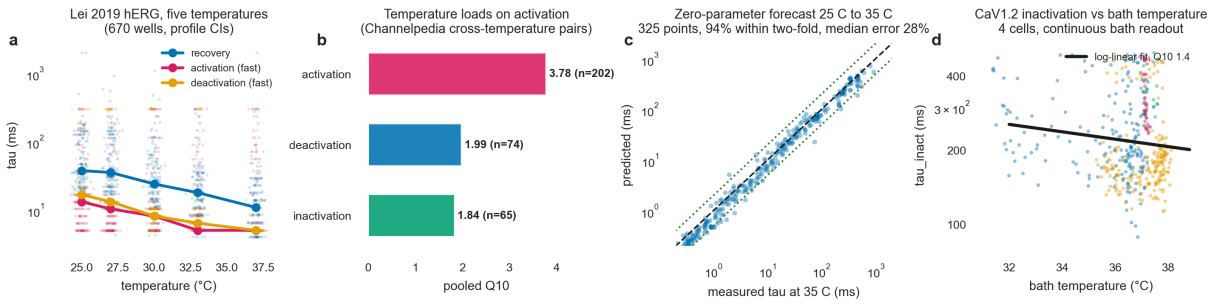


Figure 3: **Temperature moves kinetics, not midpoints.** (a) The five-temperature hERG series: per-well time constants (light points) and medians (lines) for three processes. (b) Pooled Q_{10} by process across the Channelpedia cross-temperature pairs. (c) Zero-parameter forecast of 35 °C kinetics from 25 °C tables divided by pooled Q_{10}; dotted lines mark the two-fold window. (d) $Ca_v1.2$ inactivation time constant against continuously recorded bath temperature, with the log-linear fit.

2.5 Drugs move occupancy, not kinetics

We reanalysed the five-laboratory blinded hERG dataset at waveform level with the same profile pipeline (1,750 cells contributing deactivation windows, 192,055 extracted sweep-level quantities). For each cell the deactivation time constant of tail currents under drug was referenced to the vehicle segment of the same cell, with the steady-state (highest-concentration) block as the test window. Across 970 cells with both segments valid, the median ratio of drug to vehicle time constant is 0.948 (Fig. 4a). The ratio does not drift with block depth: binned by steady-state block fraction, the stratum medians run 0.96, 0.90, 0.89, 1.01 and 1.07 from shallow to deep block (Fig. 4b), so whatever small kinetic modulation exists is bounded, does not scale with occupancy, and no drug doubles or halves the time constant. Per-laboratory medians are 1.05, 0.99, 0.81, 0.92 and 1.17; the laboratory-3 value corresponds to a 19% acceleration and is the largest deviation in the dataset. The blinded arm reproduces the unblinded one through the same frozen pipeline (medians 0.87 against 0.96; Fig. 4c), which is the internal control against analyser bias. Potency anchors are reproduced independently: our per-drug IC_{50} estimates match the official CiPA values within ten-fold for 23 of 24 drugs (Fig. 4d).

Occupancy itself is not simple one-to-one pore plugging. Three waveform-level signatures indicate state-dependent binding: Hill coefficients differ from one in 42 of 134 drug-laboratory sets; in deep block the residual current has a different shape ratio than simple scaling predicts (0.73 in laboratory 1); and the onset of block does not plateau within either exposure in 36–39% of cells. The apparent on-rate is not measurable in this dataset at all: it is flattened by solution exchange (log-log slope against concentration approximately zero), so this dataset pins equilibrium quantities and not binding kinetics, a limitation we register explicitly.

The same invariance appears outside hERG. In the Nav1.5 two-temperature series, the inactivation time constant at 35 °C is flat across the five compound levels of that study (medians 0.25–0.36 ms; Fig. 5a, inset), so the drug-independence of kinetics is not an hERG peculiarity.

2.6 The channel cards

The remaining three corpora extend the same picture beyond the Kv family (Fig. 5). Nav1.5 inactivation kinetics are fast and tight: at 35 °C the inactivation time constant runs from 0.22 to

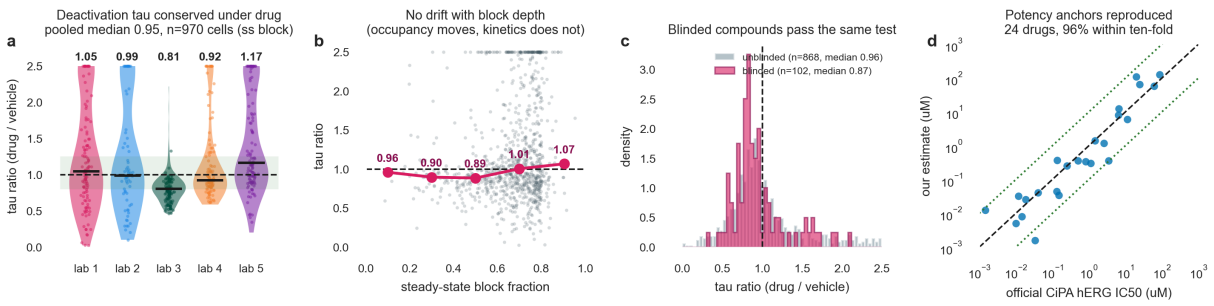


Figure 4: **Drugs move occupancy, not kinetics.** (a) Deactivation time-constant ratio, drug versus vehicle within cell, per laboratory (violins; medians marked; green band 0.80–1.25; steady-state block window). (b) The same ratio against steady-state block fraction, with stratum medians. (c) Blinded and unblinded cells through the same frozen pipeline. (d) Potency anchors: our per-drug IC_{50} estimates against the official CiPA values; dotted lines mark ten-fold.

0.37 ms across the voltage range, four cells agreeing within a narrow band, and the 25 °C series sits systematically slower as the temperature law requires (Fig. 5a). $Ca_v1.2$ calcium-dependent inactivation at +47 mV has a median time constant of 195 ms across six cells with the fit ceiling registered where the carrier is Ba^{2+} (Fig. 5b). I_{Ks} is the slow counterpart of hERG: activation time constants form the characteristic bell between 0.2 and 0.6 s and deactivation sits near 450 ms across 283 recordings, with the interquartile band narrow at every voltage (Fig. 5c). In the nine manual hERG cells the deactivation ladder is reproduced cell by cell (slow component 3.6 s at −40 mV, fast component near 10 ms), while the activation midpoints of the same nine cells spread from −12.6 to −5.4 mV (Fig. 5d): at single-cell resolution, kinetics are the shared coordinate and the midpoint is the personal one.

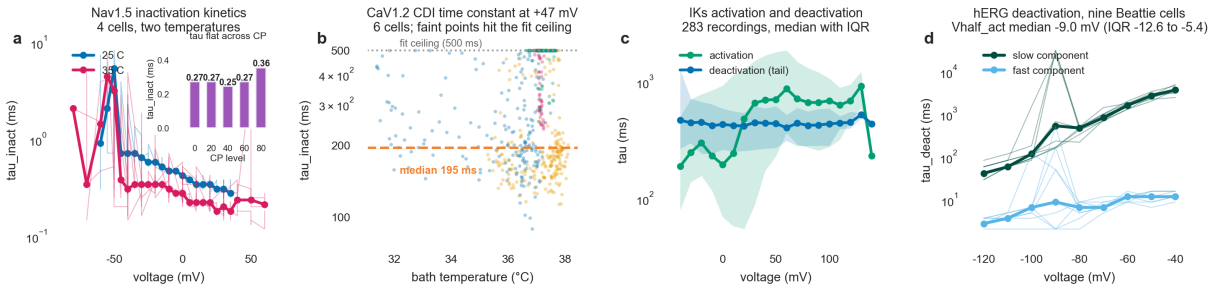


Figure 5: **Channel cards from the unified table.** (a) Nav1.5 inactivation time constant versus voltage at two temperatures (thin lines, individual cells; heavy lines, medians). Inset: the same quantity across the five compound levels of the study at 35 °C. (b) $Ca_v1.2$ inactivation time constant at +47 mV against bath temperature, six cells; faint points are flagged at the fit ceiling, and the dashed line is the median of quality-passed sweeps. (c) I_{Ks} activation and deactivation time constants, medians with interquartile bands across 283 recordings. (d) hERG deactivation, two exponential components, nine cells (thin) and medians (heavy).

2.7 Measured tables forward-predict

The point of a conserved quantity is that it can be carried into a model without refitting. We assembled channel models in the form $I = G\,m\,h\,(V - E_{\mathrm{rev}})$, with every voltage dependence a measured table and the per-cell conductance G the only free parameter, and we tested the hardest available carry: replace the sealed hERG kinetics of the assembled model by the Channelpedia population time-constant table, measured in a different laboratory on an automated rig, and

predict the manual-patch CiPA drug-screening traces cell by cell. The population-table model passes the frozen acceptance criteria in 67 of 68 cells (laboratory 3: 20/20; laboratory 4: 16/16; laboratory 5: 29/32; manual: 3/3), improving on the sealed-table model (65/68) while no cell moves from pass to fail. The two rescued cells sit in laboratory 5, whose recordings carry the largest voltage-clamp uncertainty of the panel, so the population table adds information exactly where the single-cell tables are weakest.

The assembly is compatible with the regulatory host end to end. Substituted into the O'Hara–Rudy human ventricular model in place of the published hERG and I_{Ks} formulations, the measured tables preserve action-potential morphology (APD90 difference 0.36% at 1 Hz steady state) and per-drug qNet tracking through the dynamic-binding CiPA pipeline ($\rho = 0.998$ against the official arm on the 24 shared risk-labelled drugs). A measurement-table model can therefore enter the existing scoring pipeline without changing its outputs materially, while every one of its parameters is an auditable measurement rather than a fitted constant.

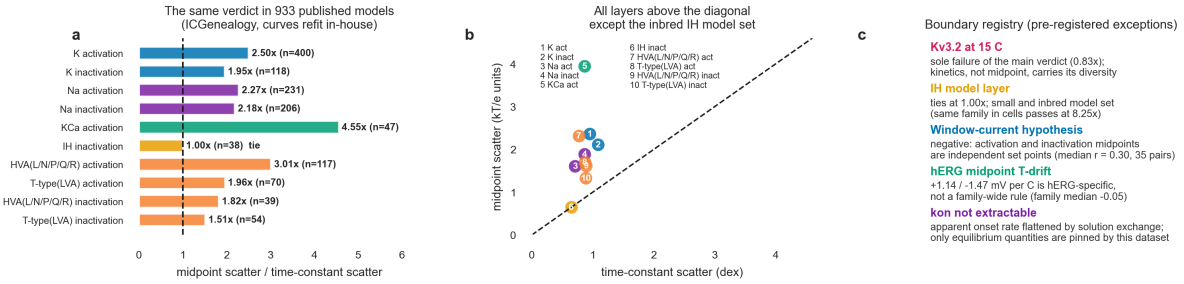


Figure 6: **The model literature and the boundary registry.** (a) The verdict ratio for ten gating layers of 933 deduplicated published models (ICGenealogy), curves refit in-house; the I_H layer ties. (b) The two scatters per layer; all layers sit above the diagonal except the inbred I_H set (numbered points keyed in panel). (c) The pre-registered exceptions and negative results, carried with the paper rather than smoothed over.

3 Discussion

A decade of identifiability work showed that the rate constants of channel models cannot be recovered from voltage-clamp data. This paper maps the complementary question, what the data do determine, at the scale the public record now allows. The answer has two parts. Structurally, the data pin time and steady state while leaving amplitude free (Eqs. (2)–(3)). Empirically, the quantities the data pin are the quantities cells share. Across seven corpora, one verdict rule finds time constants conserved and midpoints cell-specific in nearly every group, and the same asymmetry reappears in published models and across species. The two halves support each other: the modal coordinates are measurable precisely because they are constrained, and they are worth measuring because they are the stable content of the channel.

The biological reading is that a channel type is specified by its kinetics, while each cell tunes where on the voltage axis those kinetics operate. Midpoints shift with expression background, auxiliary subunits, phosphorylation and lipid state, all of which vary between cells and preparations; kinetics are set by the conformational free-energy landscape of the pore protein itself. The cross-species layer makes the same point over evolutionary rather than cell-to-cell distance.

For modelling practice the recommendation is concrete. Spend parameters on kinetic tables, which are conserved and therefore transferable, and fit midpoints per cell, which are not. A model built this way needs one free conductance per cell and predicts held-out protocols, cells and laboratories. For drug safety the corollary is that the action of a drug on a channel is, to first order, a change in occupancy tables rather than in kinetics, with a bounded kinetic correction of at most about 20% in the worst laboratory of a regulatory-scale dataset. Risk scores built on

IC_{50} alone discard the state-dependence signatures visible in the waveforms, which is where the residual risk information lives.

The boundaries are part of the result (Fig. 6c). Kv3.2 at low temperature stores its diversity in kinetics. The I_H model layer ties, and the inbred structure of that model set is the likely reason. Activation and inactivation midpoints are independent set points, not a coupled window. The apparent drug on-rate is not extractable from exchange-limited recordings. Kir and K2P channels, whose currents are leak-like or rectification-dominated rather than gated by S-shaped curves, do not enter the verdict at all, and the $Ca_v1.2$ arm rests on six cells. None of these exceptions weakens the main statement, and all of them are where the next corpora should look first.

Three limitations deserve emphasis. First, conservation is proven against cell-to-cell scatter within a preparation, not against every possible biological perturbation; disease remodelling, for example, may move kinetics as well as midpoints, and the table makes that question directly testable. Second, the verdict rule compares dimensionless scatters, and its threshold at one is a convention; the conclusions rest on the size and unanimity of the effect, not on the threshold. Third, the profile pipeline assumes the segment-local error model of each extraction, and where that model is wrong the intervals are wrong with it; the calibration panel (Fig. 1c) is the empirical check, and it currently holds within a few percent of nominal.

The master table, the extraction code and the per-layer verdict tables are public (Methods, Data availability). Every number in this paper can be regenerated from the raw waveforms by the frozen pipeline, and every extracted quantity carries its own uncertainty interval and quality flag, so the claims here can be audited row by row.

4 Methods

4.1 Data

Seven public voltage-clamp corpora and three reference layers were used (Table 1). (i) Channelpedia: automated patch-clamp recordings of ion-channel cell lines (19,282 cells; 17,987 after quality control), covering Kv, HCN, Kir, K2P and other families at 15, 25 and 35 °C, distributed as NWB files and per-cell JSON. (ii) CiPA five-laboratory hERG: manual patch-clamp recordings coordinated by HESI under the FDA Broad Agency Announcement, 28 drugs under the standardised ICH S7B Q&A 2.1 protocol with blinded replicate arms [9]; raw per-cell waveforms (62 GB, sha256-verified) at OSF osf.io/a6k5t; 1,875 cells passed trace-level quality control in our reanalysis, of which 1,750 contribute deactivation time constants to the master table. (iii) hERG five-temperature series [11] (670 wells, 25–37 °C). (iv) Nine-cell manual hERG dataset [7] (figshare doi:10.6084/m9.figshare.4702546). (v) Nav1.5 whole-cell series at 25 and 35 °C under five compound levels [12] (figshare doi:10.6084/m9.figshare.27193878). (vi) $Ca_v1.2$ pharmacology dataset with Ca^{2+} and Ba^{2+} carriers and continuous bath-temperature readout [13] (OSF g3msb). (vii) I_{Ks} activation series [14, 15] (Zenodo 8226585). Reference layers: official CiPA potency anchors for 29 drugs and seven channels; cross-species group comparisons computed from layer (i); and gating curves of 933 deduplicated published models from ICGenealogy, refit in-house.

4.2 Extraction and profile-likelihood intervals

Each protocol segment was reduced locally: mono- or bi-exponential fits to activation, inactivation, deactivation and tail segments (component count fixed per channel); Boltzmann fits to steady-state activation and availability; delays by back-extrapolation to current onset, following published good-practice guidelines [18]. No objective function couples protocols or parameters. Fit windows were set by protocol segment boundaries. Every extracted quantity carries a per-row 95% profile-likelihood interval obtained by refitting the nuisance parameters on a grid over the target parameter and applying the F-statistic threshold. Rows whose estimate reaches a fit boundary are flagged `tau_edge` and retained. The unified schema assigns source, dataset, cell, channel, family,

temperature, parameter, process, voltage, value, interval, model, quality flag, method version, drug and blinding status to every row.

4.3 Conservation verdict

For each channel-by-temperature group with at least 15 cells, the time-constant scatter is the median over voltages of the across-cell standard deviation of $\log_{10}\tau$, and the midpoint scatter is the across-cell standard deviation of $V_{1/2}$ divided by the median Boltzmann slope k. Group-level true scatter was estimated by hierarchical partial pooling with marginal likelihood, with per-row standard errors taken from the profile intervals (asymmetric half-widths averaged). The verdict ratio is midpoint scatter over time-constant scatter; a ratio above one marks the time constant as the more conserved quantity. The same rule was applied to the published-model layer after refitting each gating curve.

4.4 Temperature analysis and forecast

Pooled Q_{10} values were estimated per channel, process and voltage from cross-temperature pairs, and per process by median aggregation. The zero-parameter forecast divides the 25 °C population table by the pooled Q_{10} and compares against the measured 35 °C population table at matching channel, process and voltage.

4.5 Drug reanalysis

CiPA waveforms were segmented per cell into vehicle, drug-block and steady-state windows without using the summary metadata (segmentation was validated against the two directories that carry metadata). Tail deactivation time constants were extracted per sweep with profile intervals; the per-cell ratio references the steady-state window to the vehicle window of the same cell. Series-resistance robustness, blinded-arm concordance, concentration monotonicity, onset-rate flattening and the defect registry were computed as frozen audit stages on the same extraction.

4.6 Model assembly and host-model tests

Channel models were assembled as $I = G\,m\,h\,(V - E_{\mathrm{rev}})$ with measured tables and a per-cell conductance G as the only free parameter. The cross-laboratory test substitutes the Channelpedia population time-constant table for the sealed hERG kinetics and scores prediction of CiPA screening traces under the frozen acceptance criteria (67/68 with the population table, 65/68 with the sealed table, no pass-to-fail flips). The O'Hara–Rudy host tests kept the published $\mathrm{Na_v}1.5$ and $\mathrm{Ca_v}1.2$ formulations and replaced I_{Kr} and I_{Ks} by the measured-table implementations; APD90 was measured at 1 Hz steady state and qNet tracking followed the published dynamic-binding pipeline.

4.7 Identifiability analysis

The thirteen-parameter formulation, its calibration, and the Fisher matrix of Eq. (1) are described in the Supplementary Information. Derivatives were evaluated by central finite differences (step 10^{-3} dex for log-scaled parameters) on an independent simulation channel, with per-segment residual variance as weights. Null directions were defined as eigenvalues below 10^{-3} of the largest.

4.8 Statistics

Group verdicts are reported as pass counts with median ratios; dispersion ratios use hierarchical shrinkage as above; temperature scaling uses Arrhenius and log-linear regression; correlations are Spearman. No p-value is reported for comparisons that were not specified before the analysis. All statistics were computed from the master table by the deposited scripts.

Data and code availability

All source datasets are public (Table 1). The master table, the extraction and verdict code, the figure scripts and the per-layer verdict tables are deposited at Zenodo (https://doi.org/10.5281/zenodo.23042212) and mirrored at GitHub (https://github.com/lihuanlz/alpha-ion-channel-models). The earlier four-channel preprint is at Research Square (https://doi.org/10.21203/rs.3.rs-11218939/v1).

Competing interests

The author is employed by Shanghai Liangta Biotech Co., Ltd.

Table 1: **Data inventory.** Seven experimental corpora were re-extracted from raw waveforms by the unified profile-likelihood pipeline; three reference layers are stored in the same schema. Row and cell counts refer to the master table.

Corpus	Content	Rows	Cells
Channelpedia	74 channel types, automated patch, 15/25/35 °C	276,039	17,987
CiPA five-laboratory hERG	28 drugs, blinded arms, manual patch	192,055	1,750
Lei 2019 five-temperature	hERG, 25–37 °C, automated	10,203	670
Beattie 2018 nine-cell	hERG, six protocols, manual	207	9
Nav1.5 (27193878)	two temperatures, five compound levels	371	4
$Ca_v1.2$ (Ren 2022)	Ca^{2+}/Ba^{2+}, bath-temperature readout	562	6
I_{Ks} (Fedida/Chan)	activation series	8,147	283
Reference: CiPA anchors	official IC_{50} and Hill, 29 drugs × 7 channels	250	
Reference: cross-species	38 pairs, fold and midpoint differences	76	
Reference: ICGenealogy	933 models refit, ten gating layers	30	

Table 2: **Headline verdicts.** Ratio of midpoint scatter to time-constant scatter (shrunk estimates with profile-interval uncertainties); a ratio above one marks the time constant as the more conserved quantity.

Layer	Process	Groups	Pass	Median ratio
Channelpedia Kv	activation	64	63 (98.4%)	2.47
Channelpedia Kv	deactivation	48	43 (89.6%)	2.82
Channelpedia Kv	inactivation	35	28 (80.0%)	1.33
Channelpedia Kv	recovery	42	39 (92.9%)	2.15
Channelpedia HCN	activation (reversed)	5	5 (100%)	8.03
ICGenealogy models	ten family-by-gate layers	10	9 (90%)	2.07
Cross-species pairs	two-fold window on τ	38	38 (100%)	

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